

The Evolution of Dental Health in Dental Students at the University of Barcelona

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Abstract: The aim of this research was to study the evolution of dental health of dental students during their academic training and to assess the extent to which the knowledge acquired was reflected in their own dental care. A sample of 107 students at the schools of dentistry and medicine (the latter as a comparison group) of the University of Barcelona, Spain, underwent an oral examination and completed a questionnaire during their training in the third and fifth academic years. The oral examinations were limited to the status of teeth. Bitewing radiographs were used for both posterior sectors, and these were interpreted using the criteria proposed by Pitts (1984). The data were analyzed using the SPSS package. At the end of the study, medical students had more teeth present than dental students (29.80 vs. 28.94 [$p=0.022$]) and a lower DMFT index of 4.33 vs. 5.91 ($p=0.038$), with an FT component of 2.44 and 5.23 ($p=0.011$), respectively. All dental students underwent more treatment of all types than medical students, and dental health habits and knowledge were superior in dental students. The third year was the key year for decision making with regard to the student's dental health. We conclude that dental students are highly motivated about maintaining their dental health and their dental education experiences appear to have had a clear influence on this behavior.

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Over the last few years a significant decline in the prevalence of dental diseases such as caries has been observed in the majority of developed countries. This well-documented decline is also evident in Spain.¹ The majority of authors attribute these changes to an improvement in oral health habits and, particularly, to the widespread use of fluoridated toothpaste.² Dental health professionals have an important role in the improvement of the public's health education level. For this reason, the acquisition of knowledge and attitudes relating to dental health and the prevention, control, and treatment of dental problems during the future dentists' training period is very important.³ At present, it is assumed that the decrease in the prevalence of dental caries in many population groups is also related to a reduction in the activity and the speed of progression of the carious lesions.⁴ This has led to a change in the dentists' approach to dental care, which is more oriented toward prevention rather than restoration in order to avoid or postpone invasive treatment.⁵

The process of adopting this change in dental care attitudes regarding their own dental care must be learned and practiced throughout the dentists' learning process, especially during their undergraduate training in dental school.⁶ Therefore, a study of how dental students apply this knowledge to their own oral health care during dental school could, in fact, be of a great value since the students are the ones who will apply these same behavior patterns to their own patients in their practices. The object of this study was to evaluate the influence of undergraduate training on the dental students' oral health and on their dental care decisions.

Population and Methods

The study population was all third-year students at the dental and medical schools of the University of Barcelona, Spain. The participants gave their prior

Table 1. Study demographics

	Dentistry	Medicine	Total
Initial sample	55	52	107
Lost by age	4	0	4
Lost in 2 nd examination	4	7	11
Final sample	47	45	92
Females	28	28	60
Males	15	17	32
Mean initial age	21.1	20.5	20.8
Mean final age	23.0	22.8	22.9

consent for study participation and were examined on two occasions during the study—first in the third year and again before graduating in their fifth year. The initial data were obtained at the start of the 1993-94 academic year and the final data during the middle of the 1995-96 academic year.

During the first examination, 107 subjects of both sexes were examined, accounting for 60 percent of the third-year dental and medical students. After the first examination, four subjects (aged between twenty-eight and forty years) were withdrawn from the study in order to obtain a homogeneous sample as far as age was concerned. In the second examination, eleven subjects were lost, so that the final study sample was made up of ninety-two students: forty-seven from dentistry and forty-five from medicine (Table 1).

The examinations were carried out at the clinic in the School of Dentistry under standardized conditions by two trained, calibrated examiners. Bitewing X-rays were taken (31 x 41 mm, D sensitivity) for both posterior quadrants of the mouth, using the Klauser Kwik-Bite positioner. The oral examination was limited to the teeth and the treatment, if any, they received. For recording caries ("D" component of the DMF), the basic WHO 1987 criteria were used.⁷ The initial or supposedly inactive lesions that did not fulfil the WHO criteria were included in a new category called "initial/remineralized caries lesion." The x-rays were interpreted in accordance with the criteria proposed by Pitts in 1983.⁸

Each student replied anonymously to a self-administered questionnaire composed of seventeen questions on habits, knowledge, and attitude regarding dental care. The questionnaire was a modification of the Cavaillon et al.⁹ and Cuenca et al.¹⁰ questionnaires and contained questions pertinent to this study.

The data obtained were introduced into a database and analyzed with an SPSS personal computer statistics package. Frequency distribution measures were used for the qualitative variables and measures

of central tendency and dispersion for the quantitative variables. The chi-squared test was used to compare qualitative variables, significance of difference between means was determined by t-test, and 95 percent confidence interval was calculated. In all cases, a p value of 0.05 was used for the level of significance.

Results

The final study sample was made up of ninety-two students—forty-seven from dentistry and forty-five from medicine—with mean ages of 21.1 and 20.5 years respectively. Women accounted for 68 percent and 62 percent of the study population, respectively (Table 1).

Basic Dental Health Indices at Baseline and Final

At the start, both groups exhibited a similar mean for the number of teeth present, healthy untreated teeth, and percent DMF. At the end of the study, the number of teeth present and healthy untreated teeth was lower in dental students, 28.94 vs. 29.80 ($p=0.022$) and 21.34 vs. 23.69 ($p=0.009$), respectively. (See Table 2.)

The mean DMFT at the start was higher in dental students: 5.04 vs. 3.38 ($p=0.010$); this was also the case at the end: 5.91 vs. 4.33 ($p=0.038$). Regarding the DMFT components, the mean active caries at the start were similar in both groups, but the filled component (FT), 4.11 vs. 2.04 ($p=0.001$), was significantly higher in dental students. At the end of the study, the DT index showed that the medical students presented more active untreated caries than at the start, while the opposite situation existed among dental students. The restoration index (RI) was clearly higher among dental students.

Types of Treatment Received

At the start of the study, dental students underwent more treatment of all types, more amalgams, composites, crowns, teeth with endodontic treatment, and more extracted third molars than their medical counterparts. At the end of the study, the differences in treatment received were statistically significant for amalgams ($p=0.004$) and third molars extracted ($p=0.000$), with the dental students receiving more treatment in both categories. (See Table 3.)

Table 2. Dental health indices at baseline and final examination

	Mean	S.D.	C.I. 95%	Level of significance*
TEETH PRESENT				
Baseline				
Dentistry	29.02	±1.80	28.49; 29.45	0.419
Medicine	28.89	±2.21	28.23; 29.55	
Final				
Dentistry	28.94	±1.85	28.39; 29.48	0.022
Medicine	29.80	±1.71	29.28; 30.31	
SOUND AND UNTREATED TEETH				
Baseline				
Dentistry	22.64	±3.78	21.53; 23.75	0.429
Medicine	23.29	±4.08	22.06; 24.51	
Final				
Dentistry	21.34	±4.24	20.09; 22.59	0.009
Medicine	23.69	±4.16	22.44; 24.94	
EARLY/REMINERALIZED CARIES LESIONS				
Baseline				
Dentistry	1.79	±1.93	1.22; 2.35	0.055
Medicine	2.78	±2.84	1.92; 3.63	
Final				
Dentistry	2.49	±1.86	1.94; 3.04	0.634
Medicine	2.27	±2.57	1.49; 3.04	

	%DMFT>0	DT Mean±SD	MT Mean±SD	FT Mean±SD	DMFT (CI 95%)	DMFS (CI 95%)	RI
Baseline							
Dentistry	83%	0.89 ±1.63	0.04 ±0.20	4.11 ±3.02	5.04 (4.08; 6.01)	7.83 (5.99; 9.67)	81.4%
Medicine	82.2%	1.18 ±1.47	0.16 ±0.37	2.04 ±2.72	3.38 (2.55; 4.21)	5.40 (4.00; 6.97)	57.6%
Level of significance	0.924**	0.383*	0.074*	0.001*	0.010*	0.051*	—
Final							
Dentistry	87.2%	0.60 ±1.15	0.08 ±0.28	5.23 ±3.71	5.91 (4.76; 7.06)	10.11 (7.81; 12.41)	88.5%
Medicine	91.1%	1.67 ±1.72	0.22 ±0.47	2.44 ±2.64	4.33 (3.36; 5.30)	7.11 (5.10; 9.13)	56.4%
Level of significance	0.550**	0.01*	0.097*	0.000*	0.038*	0.052*	—

* Level of significance for t-test and **chi-square test.

Dental Health Habits and Perception

The brushing frequency at the start of the study was clearly higher in dental students than in medical students, and this was also the case at the end: 63.8 percent vs. 37.8 percent ($p=0.044$). Also, the average duration of tooth brushing was greater in dental students and gradually increased during the study. There was a similar finding with respect to the use of dental floss, which dental students used to a greater extent than medical students: 80.9 percent vs. 22.2 percent ($p=0.000$). (See Table 4.)

The perception of the state of oral health was more positive among dental students than among their medical peers, and this increased throughout the study. In addition, 46.8 percent of dental students

stated that they had changed their dentist during their university studies, compared to just 17.8 percent of medical students ($p=0.003$).

Knowledge, Attitudes, and Decision Making Regarding Dental Health

The majority of dental students considered a regular six-month (29.5 percent) or yearly (38.6 percent) checkup as important, whilst the majority of medical students considered that it depended on the level of pathology (55.8 percent). At the end of the study, opinions were reversed: 66.0 percent of the dental students considered that regular checkups should be scheduled depending on the level of pathology

Table 3. Types of treatments, number (N), and mean $\pm$ SD of teeth with restorative treatment, endodontic treatment, symptomatic teeth, and extracted third molar

	Amalgams (N) Mean	Composites (N) Mean	Crowns (N) Mean	Endodontics (N) Mean	Symptomatic Teeth (N) Mean	3 rd Molar Extracted (N) Mean
Baseline						
Dentistry	(138) 2.94 $\pm$ 2.82	(54) 1.15 $\pm$ 2.18	(11) 0.23 $\pm$ 0.56	(21) 0.45 $\pm$ 0.83	(2) 0.04 $\pm$ 0.20	(24) 0.51 $\pm$ 1.10
Medicine	(70) 1.56 $\pm$ 2.49	(21) 0.47 $\pm$ 1.59	(6) 0.13 $\pm$ 0.41	(11) 0.24 $\pm$ 0.68	(2) 0.11 $\pm$ 0.53	(7) 0.16 $\pm$ 0.52
Level of significance*	0.015	0.089	0.324	0.203	0.422	0.051
Final						
Dentistry	(154) 3.28 $\pm$ 2.95	(86) 1.83 $\pm$ 2.89	(20) 0.43 $\pm$ 0.88	(23) 0.49 $\pm$ 0.88	(11) 0.23 $\pm$ 0.56	(63) 1.34 $\pm$ 1.45
Medicine	(73) 1.62 $\pm$ 2.35	(38) 0.84 $\pm$ 1.85	(10) 0.22 $\pm$ 0.70	(11) 0.24 $\pm$ 0.57	(10) 0.22 $\pm$ 0.47	(13) 0.29 $\pm$ 0.84
Level of significance*	0.004	0.054	0.223	0.116	0.913	0.000
Total number of new treatments performed during the study						
Dentistry	16	32	9	2	9	39
Medicine	3	17	4	0	8	6
Total	19	49	13	2	17	45

* Level of significance for t-test.

Table 4. Dental health habits

Times	BASELINE		FINAL	
	Dentistry (N)%	Medicine (N)%	Dentistry (N)%	Medicine (N)%
1. How many times a day do you brush your teeth?				
<1 time	(0) 0%	(4) 9.3%	(0) 0%	(3) 6.7%
1 time	(5) 11.4%	(17) 39.5%	(7) 14.9%	(11) 24.4%
2 times	(17) 38.6%	(13) 30.2%	(10) 21.3%	(14) 31.1%
3 times	(22) 50%	(9) 20.9%	(30) 63.8%	(17) 37.8%
Level of significance*	0.001		0.044	
2. How many minutes do you spend each time?				
1 Minute	(3) 6.8%	(10) 23.3%	(2) 4.3%	(2) 4.4%
1 – 2 Minutes	(24) 54.5%	(23) 53.4%	(20) 42.6%	(25) 55.6%
>2 Minutes	(17) 38.6%	(10) 23.3%	(25) 53.2%	(18) 40%
Level of significance*	0.061		0.438	
3. Do you usually use dental floss?				
Yes	(34) 77.3%	(8) 18.6%	(38) 80.9%	(10) 22.2%
No	(10) 22.7%	(35) 81.4%	(9) 19.1%	(35) 77.8%
Level of significance*	0.000		0.000	
4. How do you value your dental health?				
Good/very good	(33) 75%	(26) 60.4%	(37) 78.7%	(28) 62.2%
Bad/very bad	(6) 13.6%	(14) 32.6%	(9) 19.1%	(13) 28.9%
I don't know	(5) 11.4%	(3) 7%	(1) 2.1%	(4) 8.9%
Level of significance*	0.159		0.232	
5. When do you go to the dentist?				
In acute phase	(8) 18.2%	(27) 62.8%	(4) 8.5%	(23) 51.1%
To check up	(36) 81.8%	(16) 37.2%	(43) 91.5%	(22) 48.9%
Level of significance*	0.000		0.000	
6. Have you changed your usual dentist during the last five years?*				
Yes	—	—	(22) 46.8%	(8) 17.8%
No	—	—	(25) 53.2%	(37) 82.2%
Level of significance*	—		0.003	

* Level of significance for chi-square test

** Question asked at final examination

($p=0.000$). Faced with an incipient caries lesion, the majority of dental students (90.9 percent) considered that active preventive measures should be instituted; a small percentage (9.1 percent) said that it should be filled “as soon as possible.” At the end of the study, no dental student felt the need to fill an incipient lesion “as soon as possible.” (See Table 5.)

All dental students considered that their university studies had helped them make decisions regarding their own dental treatment, and the key academic year for this awareness was the third year (63.8 percent), followed by the fourth (17 percent). (See Table 6.) Also, 38.3 percent of these students had changed some of their fillings in the last five years, compared to just 11.1 percent of medical students ($p=0.026$). With regard to the decision to change a filling, dental students, in consultation with their dentist, took the initiative to a greater extent than did medical students: 44.7 percent vs. 31.1 percent ($p=0.497$).

Discussion

These results show that, with regard to restorative dentistry as well as third molar exodontia, the dental treatment undergone by dental students was far greater than that of their medical peers, even though disease prevalence was not greater in these students. The results also show that dental students possessed a very high oral health care level and their motivation and perception were very positive. Further, the majority had changed their dentist and had numerous fillings replaced. To a large extent, they influenced the decisions for the treatment of their teeth, and their dental training appeared to have had a decisive influence in this behavior. While with sev-

eral parameters, e.g., oral hygiene practices, there were significant differences between the two groups (better among dental students), some oral health indices were not better than those of their medical peers. In fact, dental students completed their career with a lower number of teeth present ($p=0.022$), a lower number of healthy untreated teeth ($p=0.009$), and a higher mean DMFT ($p=0.038$).

With regard to the caries history, when the dental and medical students were in their third academic year, they showed the same DMF prevalence and the same means for active carious lesions (D component) although the DMFT and DMFS indices were significantly higher among dental students. At the end of the study, the DMFT index had increased in both groups, although it continued to be higher in dental students primarily due to the filled component ($p=0.000$). Moreover, the analysis of the relative risk showed that while the risk of a new surface with caries appearing was slightly higher in medical students, the risk of a new filled surface appearing was slightly more than twice as high in dental students.¹¹ In addition, the analysis of the types of treatment showed that dental students had more third molars extracted (thirty-nine vs. six), amalgams (sixteen vs. three), composites, crown placements, and endodontic treatment. By far, dental students received more dental treatment than that of their medical peers.

There appears to be somewhat of a contradiction between acquired knowledge and its practice among dental students. For example, no dental student considered that incipient carious lesions should be filled as soon as possible, and 66 percent considered that the frequency of checkups should be related to the risk level. Therefore, there is no concordance between what they consider should be a regime

Table 5. Attitudes regarding dental health

Time	BASELINE		FINAL	
	Dentistry (N)%	Medicine (N)%	Dentistry (N)%	Medicine (N)%
1. The regular checkup must be:				
Every six months	(13) 29.5%	(6) 14%	(10) 21.3%	(9) 20%
Once a year	(17) 38.6%	(13) 30.2%	(6) 12.8%	(23) 51.1%
Depends on individual's pathology	(14) 31.8%	(24) 55.8%	(31) 66%	(13) 28.9%
Not necessary	(0) 0%	(0) 0%	(0) 0%	(0) 0%
Level of significance*	0.057		0.000	
2. In your opinion, the correct attitude when faced with an incipient caries lesion would be:				
Wait and see its progress	(0) 0%	(0) 0%	(1) 2.1%	(1) 2.2%
Take preventive measures	(40) 90.9%	(32) 74.4%	(46) 97.9%	(40) 88.9%
Fill as soon as possible	(4) 9.1%	(11) 25.6%	(0) 0%	(4) 8.9%
Level of significance*	0.042		0.112	

Table 6. Decision making regarding dental health

FINAL**		
Course	Dentistry (N) %	Medicine (N) %
1. Do you think your current education has influenced the decisions you have taken for your own dental treatment? Can you determine when this influence began?		
1 st	(4) 8.5%	(3) 6.7%
2 nd	(4) 8.5%	(0) 0%
3 rd	(30) 63.8%	(7) 15.6%
4 th	(8) 17%	(1) 2.2%
5 th	(1) 2.1%	(5) 11.1%
No	(0) 0%	(29) 64.4%
Level of significance*	0.000	
2. Have you changed any fillings during the last few years. If so, why?		
Cosmetics	(2) 4.3%	(0) 0%
Break or failure	(4) 8.5%	(3) 6.7%
Suspect caries relapse	(7) 14.9%	(1) 2.2%
Caries relapse	(5) 10.6%	(1) 2.2%
I haven't changed any filling	(29) 61.7%	(40) 88.9%
Level of significance*	0.026	
3. The fillings made during the last few years in your mouth were decided by:		
My dentist	(12) 25.5%	(10) 22.2%
My usual dentist and me	(18) 38.3%	(13) 28.9%
Another dentist and me	(3) 6.4%	(1) 2.2%
No new fillings	(11) 23.4%	(17) 37.8%
Other reasons	(3) 6.4%	(4) 8.9%
Level of significance*	0.497	
*Level of significance for chi-square test.		
**Questions asked at final examination		

to control the infection and what they actually carry out in practice on their own teeth.

In conclusion, in light of these results it can be said that: 1) dental students in this study were highly motivated and had a good perception of their own dental health; 2) they received all types of dental treatment during the university training; and 3) their dental education may have influenced their initiative in seeking and obtaining this treatment.

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